

THERE ARE NO RIESZ BASES OF EXPONENTIALS IN BALLS AND TRIANGLES

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ABSTRACT. We prove that $L^2(\Omega)$, where Ω is a ball in $\mathbb{R}^n$, $n \geq 2$, or a nondegenerate triangle in $\mathbb{R}^2$, admits no Riesz basis of exponentials. Using Beurling weak limits of translates of the frequency set, we construct a stationary system carrying spectral cutoff projections and comparison projections built from smooth bumps. The Riesz basis assumption forces the two projections to remain uniformly at operator-norm distance less than one. The comparison family is continuous in operator norm, while boundary geometry produces two pointwise limits of the spectral cutoffs whose ranges are strictly nested. Both limiting projections would then lie at distance less than one from the same comparison projection, which is impossible.

1. INTRODUCTION AND MAIN RESULT

Fix an integer $n \geq 2$. Let $\Omega \subset \mathbb{R}^n$ be an open ball and, for $\lambda \in \mathbb{R}^n$, write

$$e_\lambda(x) = e^{2\pi i \lambda \cdot x}.$$

We ask whether there can be a sequence $\Lambda \subset \mathbb{R}^n$ for which $\{e_\lambda\}_{\lambda \in \Lambda}$ is a Riesz basis of $L^2(\Omega)$. Thus the synthesis map

$$(c_\lambda) \mapsto \sum_{\lambda \in \Lambda} c_\lambda e_\lambda$$

is an isomorphism from $\ell^2(\Lambda)$ onto $L^2(\Omega)$; equivalently, there are constants $0 < A \leq B < \infty$ such that

$$A \sum_{\lambda \in \Lambda} |c_\lambda|^2 \leq \left\| \sum_{\lambda \in \Lambda} c_\lambda e^{2\pi i \lambda \cdot x} \right\|_{L^2(\Omega)}^2 \leq B \sum_{\lambda \in \Lambda} |c_\lambda|^2 \quad (1.1)$$

for every finitely supported family (c_λ) , and the exponential system is complete.

The existence of an exponential Riesz basis for Euclidean balls is a longstanding folklore problem in harmonic analysis. Bruna, Massaneda

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and Ortega-Cerdà [2, Section 5, p. 350] explicitly formulated the conjecture that balls in dimensions greater than one have no such basis. The planar case was still recorded as open in 2025 [13, Section 1.2].

The main result is the following.

Theorem 1.1. *For every $n \geq 2$, no ball of positive radius in $\mathbb{R}^n$ admits a Riesz basis of exponentials.*

An invertible affine change of variables gives the same conclusion for all nondegenerate ellipsoids in $\mathbb{R}^n$; see Section 8. The argument also excludes exponential Riesz bases on every nondegenerate triangle in $\mathbb{R}^2$ (Theorem 7.1). More generally, Remark 6.4 states a boundary-measure criterion under which the same projection obstruction applies to other open sets.

The orthogonal version of this problem was initiated by Fuglede [4]. Iosevich, Katz and Pedersen [7] proved that balls in dimensions at least two admit no orthogonal basis of exponentials. Quantitative restrictions on orthogonal exponential systems on a planar ball were obtained by Iosevich and Kolountzakis [8], and subsequently strengthened by Zakharov [16]. These results concern pairwise orthogonality. For Riesz bases, Kozma, Nitzan and Olevskii [12] constructed a bounded subset of the real line without an exponential Riesz basis and obtained a quantitative lower bound for the Riesz constant of a hypothetical basis on a planar ball, without excluding its existence.

Positive results are known for polygonal domains. Lyubarskii and Rashkovskii [14] constructed exponential Riesz bases for centrally symmetric convex polygons whose vertices lie on a lattice using entire functions of two complex variables. Debernardi and Lev [3] later removed the arithmetic restriction, proving existence for every centrally symmetric convex polygon and, more generally, for centrally symmetric convex polytopes whose faces of all dimensions are centrally symmetric.

A related construction principle is lattice multi-tiling: the lattice translates of a bounded measurable domain cover almost every point the same finite number of times. Grepstad and Lev [5] proved that such domains admit exponential Riesz bases under a Riemann measurability assumption; Kolountzakis [9] gave a simpler proof and removed this regularity assumption. Kozma and Nitzan proved existence for finite unions of intervals [10] and for finite unions of rectangles with sides parallel to the coordinate axes [11].

There is a function-theoretic reformulation of the problem. If

$$PW_\Omega = \left\{ F(z) = \int_\Omega f(x) e^{-2\pi i z \cdot x} dx : f \in L^2(\Omega) \right\}, \quad z \in \mathbb{C}^n,$$

with the norm transported from $L^2(\Omega)$, then PW_Ω is a Hilbert space of entire functions. For real λ ,

$$F(\lambda) = \langle f, e_\lambda \rangle_{L^2(\Omega)}.$$

Hence Λ is the frequency set of an exponential Riesz basis precisely when the restriction map

$$F \longmapsto (F(\lambda))_{\lambda \in \Lambda}$$

is an isomorphism from PW_Ω onto $\ell^2(\Lambda)$. This map is the adjoint of synthesis under the Fourier–Laplace identification. In other words, the problem is to clarify the existence of complete interpolating sequences for a Paley–Wiener space in n variables.

The proof relies on a geometric feature of the ball: a nontrivial translate of its boundary meets the original boundary in a set of $(n-1)$ -dimensional surface measure zero.

The proof has four steps. First, separation of the frequency set allows us to replace its points by orthonormal translates of a small smooth bump. The common Fourier factor is invertible on the bounded domain, so the basis property gives an exact estimate $\|P - Q_\Lambda\| < 1$ between two orthogonal projections on $L^2(\mathbb{R}^n)$. Second, we take weak limits of real translates of Λ and construct an invariant probability measure on the resulting compact space. This use of weak limits originates in Beurling’s work on sampling and interpolation [1]; a modern account of the topology is given in [6, Section 4]. Third, scalar Bochner measures and continuous box averages give spectral cutoffs Π_t and comparison projections M_t on the stationary space, with $\|\Pi_t - M_t\| \leq \gamma < 1$ for almost every t . The averaging estimate is proved by applying the original projection bound to compactly supported functions and integrating their scalar inner products. Finally, a boundary crossing gives two limiting projections with a strict inclusion of their ranges. Norm-continuity of M_t places both at distance at most γ from M_{t_0} , a contradiction.

Sections 2–5 establish the analytic construction for any bounded open domain with boundary of Lebesgue measure zero. Section 6 supplies the geometric step for balls, and Sections 7 and 8 give the other applications.

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The Lean formalization is available at [RieszEuclidean](#). The repository contains the proof blueprint, the Lean sources, a standalone proof file, and the verification records. All blueprint obligations are proved, and the public statements have been checked by Comparator and Lean’s kernel.

2. FROM COMPLETE INTERPOLATION TO BUMP PROJECTIONS

Throughout Sections 2–5, Ω is a bounded open subset of $\mathbb{R}^n$ of positive measure. Inner products are linear in the first variable. We use the

unitary Fourier transform

$$(\mathcal{F}a)(x) = \int_{\mathbb{R}^n} a(\xi) e^{2\pi i \xi \cdot x} d\xi,$$

initially defined on $L^1 \cap L^2$, and set

$$P = \mathcal{F}^{-1} M_{\mathbf{1}_\Omega} \mathcal{F}, \quad p(y) = \int_{\Omega} e^{-2\pi i y \cdot x} dx. \quad (2.1)$$

Thus P is an orthogonal projection on $L^2(\mathbb{R}^n)$ and commutes with translations. For compactly supported $a \in L^2$, Fourier inversion and Fubini give $(Pa)(v) = \int p(v - w)a(w) dw$ almost everywhere. Indeed $a \in L^1$, Ω has finite measure and p is bounded, so this formula follows directly by interchanging the integrals defining P .

2.1. Distance between orthogonal projections. For orthogonal projections P and Q on the same Hilbert space, their operator-norm distance is $\|P - Q\|$. We use this distance throughout.

Lemma 2.1. *Let P and Q be orthogonal projections on a Hilbert space. The map*

$$P : \text{Ran } Q \longrightarrow \text{Ran } P$$

is a bounded isomorphism if and only if

$$\|P - Q\| < 1. \quad (2.2)$$

Proof. Assume first that $\|P - Q\| = \gamma < 1$. If $u \in \text{Ran } Q$, then

$$\|(I - P)u\| = \|(Q - P)u\| \leq \gamma \|u\|,$$

and therefore

$$\|Pu\|^2 = \|u\|^2 - \|(I - P)u\|^2 \geq (1 - \gamma^2)\|u\|^2.$$

This gives injectivity and a lower bound. To obtain surjectivity, put $A = P|_{\text{Ran } Q}$ and $B = Q|_{\text{Ran } P}$. On the closed subspace $\text{Ran } P$,

$$I - AB = P(P - Q).$$

Thus $\|I - AB\| \leq \gamma < 1$, so the Neumann series makes AB invertible. It follows that A is onto, and the lower bound makes its inverse bounded.

Conversely, suppose $A = P|_{\text{Ran } Q}$ is an isomorphism. Set $K = 1 + \|A^{-1}\|$ and $c = K^{-1}$, so $0 < c \leq 1$, including when both ranges are zero. For $u \in \text{Ran } Q$ we have $\|Au\| \geq c\|u\|$. For $y \in \text{Ran } P$, take $u = A^{-1}y$. Orthogonality gives

$$\|y\|^2 = \langle u, y \rangle = |\langle u, Qy \rangle| \leq \|u\| \|Qy\| \leq K \|y\| \|Qy\|.$$

For $y \neq 0$ we may cancel $\|y\|$; for $y = 0$ the conclusion is immediate. Hence $\|Qy\| \geq c\|y\|$. Pythagoras now gives

$$\|(I - P)Q\| \leq \sqrt{1 - c^2} < 1, \quad \|(I - Q)P\| \leq \sqrt{1 - c^2} < 1.$$

For every x ,

$$(P - Q)x = P(I - Q)x - (I - P)Qx,$$

and the two terms on the right are orthogonal. Since $(I - Q)x$ and Qx are also orthogonal,

$$\|(P - Q)x\|^2 \leq \max\{\|P(I - Q)\|^2, \|(I - P)Q\|^2\} \|x\|^2.$$

Now $\|P(I - Q)\| = \|(I - Q)P\|$, and both numbers are strictly smaller than one. This proves (2.2). $\square$

We shall also use the following immediate consequence.

Lemma 2.2. *Let R_- and R_+ be orthogonal projections with*

$$\text{Ran } R_- \subsetneq \text{Ran } R_+.$$

There is no orthogonal projection M such that

$$\|R_- - M\| < 1, \quad \|R_+ - M\| < 1.$$

Proof. By Lemma 2.1, M maps each of the two ranges isomorphically onto $\text{Ran } M$. Choose $w \in \text{Ran } R_+ \setminus \text{Ran } R_-$. Surjectivity on the smaller range gives $u \in \text{Ran } R_-$ with $Mu = Mw$. Then $0 \neq w - u \in \text{Ran } R_+$ and $M(w - u) = 0$, contradicting injectivity on $\text{Ran } R_+$. $\square$

2.2. Separated frequencies and an exact bump representation.

Lemma 2.3. *Suppose that $\{e_\lambda : \lambda \in \Lambda\}$ is a Riesz basis of $L^2(\Omega)$. There are $\delta, r > 0$, a real nonnegative $b \in C_c^\infty(\mathbb{R}^n)$ and $\gamma < 1$ such that Λ is δ -separated,*

$$\|b\|_2 = 1, \quad \text{supp } b \subset B(0, r), \quad 2r < \delta, \quad \inf_{x \in \Omega} |\mathcal{F}b(x)| > 0, \quad (2.3)$$

and the orthogonal projection Q_Λ onto the closed span of $\{b(\cdot - \lambda) : \lambda \in \Lambda\}$ satisfies

$$\|P - Q_\Lambda\| \leq \gamma < 1. \quad (2.4)$$

Proof. Apply the lower Riesz bound to coefficients 1 and -1 at distinct frequencies. Since $|e^{iu} - e^{iv}| \leq |u - v|$,

$$2A \leq \|e_\lambda - e_\eta\|_{L^2(\Omega)}^2 \leq 4\pi^2 |\lambda - \eta|^2 \int_\Omega |x|^2 dx.$$

The integral is finite and positive. This gives a positive separation constant δ and, by disjoint balls of radius $\delta/2$, a uniform bound on the number of frequencies in any ball of fixed radius.

Choose a real nonnegative smooth compactly supported b_0 with $\|b_0\|_2 = 1$. For small $\varepsilon > 0$ put $b(\xi) = \varepsilon^{-n/2} b_0(\xi/\varepsilon)$. Its support is as small as required and

$$(\mathcal{F}b)(x) = \varepsilon^{n/2} (\mathcal{F}b_0)(\varepsilon x).$$

Since $(\mathcal{F}b_0)(0) > 0$ and $\overline{\Omega}$ is compact, this proves (2.3) for a sufficiently small fixed ε .

The translates $b_\lambda = b(\cdot - \lambda)$ have disjoint supports and norm one. Consequently

$$W_\Lambda : \ell^2(\Lambda) \longrightarrow \text{Ran } Q_\Lambda, \quad W_\Lambda c = \sum_{\lambda \in \Lambda} c_\lambda b_\lambda,$$

is unitary. For finitely supported c ,

$$\mathcal{F}(PW_\Lambda c) = \mathbf{1}_\Omega(\mathcal{F}b) \sum_{\lambda \in \Lambda} c_\lambda e_\lambda. \quad (2.5)$$

Both sides extend continuously to $\ell^2(\Lambda)$. On the right, synthesis is an isomorphism onto $L^2(\Omega)$ and multiplication by $\mathcal{F}b$ is an isomorphism of that space. Hence $P : \text{Ran } Q_\Lambda \rightarrow \text{Ran } P$ is an isomorphism. Lemma 2.1 gives (2.4). $\square$

Fix b, δ, r and γ from this lemma. For every δ -separated set Γ , let Q_Γ be the orthogonal projection onto the closed span of the corresponding bumps. Explicitly,

$$Q_\Gamma a = \sum_{\lambda \in \Gamma} \langle a, b_\lambda \rangle b_\lambda. \quad (2.6)$$

The bumps represent the original synthesis map exactly.

3. BEURLING WEAK LIMITS AND A STATIONARY SYSTEM

3.1. Weak convergence of separated sets. We use Beurling's notion of weak convergence of point sets [1]. For a modern treatment see [6, Section 4]. A sequence of closed sets Γ_j converges weakly to Γ if, for every $R, \varepsilon > 0$, eventually

$$\Gamma_j \cap B(0, R) \subset \Gamma + B(0, \varepsilon), \quad \Gamma \cap B(0, R) \subset \Gamma_j + B(0, \varepsilon). \quad (3.1)$$

For sets with a common positive separation constant, this is equivalent to vague convergence of their counting measures $\eta_\Gamma = \sum_{\lambda \in \Gamma} \delta_\lambda$. To see this, weak convergence matches the finitely many points on each compact region, and continuous compactly supported test functions give vague convergence. Conversely, testing on small disjoint balls around limit points forces the matching points to exist. A sequence of unmatched points in a fixed compact set would have a convergent subsequence and would be detected by another nonnegative continuous test function. Uniform separation prevents more than one point from approaching a limit point.

Lemma 3.1. *The space of δ -separated subsets of $\mathbb{R}^n$, including the empty set, is compact and metrizable in the topology (3.1). Translations act jointly continuously (in the translation and in the subset) on this space.*

Proof. The separation bound gives a uniform bound on the mass of the counting measures on each compact set. A diagonal extraction against a countable dense collection of continuous compactly supported functions

gives a vaguely convergent subsequence from any sequence of these measures. Its limit is again the counting measure of a δ -separated set: on a ball of diameter less than δ there is at most one point, and a point either converges along a subsequence or leaves each compact subset. Applying this observation on a countable cover identifies the limit and rules out both fractional masses and multiplicities. The same countable test functions metrize vague convergence on this uniformly locally bounded family. Sequential compactness thus implies compactness. If $\Gamma_j \rightarrow \Gamma$ and $y_j \rightarrow y$, the local matching criterion immediately gives $\Gamma_j - y_j \rightarrow \Gamma - y$. $\square$

Let

$$X = \overline{\{\Lambda - y : y \in \mathbb{R}^n\}}, \quad T_y \Gamma = \Gamma - y.$$

This is a compact metrizable translation-invariant space. Its elements are precisely the weak limits of sequences of translates of Λ .

3.2. The projection estimate survives weak limits. We call a sequence of operators A_j pointwise convergent to A if $\|A_j f - A f\| \rightarrow 0$ for every fixed vector f . We use the same convention for operator families depending on a real parameter.

Lemma 3.2. *If $\Gamma_j \rightarrow \Gamma$ in X , then $Q_{\Gamma_j} \rightarrow Q_\Gamma$ pointwise on $L^2(\mathbb{R}^n)$. In particular,*

$$\|P - Q_\Gamma\| \leq \gamma \quad (\Gamma \in X). \quad (3.2)$$

Proof. First take a compactly supported $a \in L^2(\mathbb{R}^n)$. Choose a ball containing the closed r -neighborhood of its support in its interior and having no point of Γ on its boundary. Within this ball, weak convergence and separation give, for large j , a bijection between the finitely many centers of Γ_j and Γ , with matched centers converging. Only centers in this ball can contribute to (2.6). Translations of b are continuous in L^2 , so each term of this finite sum converges in L^2 . Thus $Q_{\Gamma_j} a \rightarrow Q_\Gamma a$. Compactly supported functions are dense, and all these projections have norm at most one, proving pointwise convergence.

Conjugating by translations preserves P and sends Q_Λ to the projection for the translated set. Thus every translate satisfies (2.4). Taking a pointwise limit and testing on an arbitrary vector gives (3.2). $\square$

Since P is nonzero, the empty set cannot belong to X : its projection is zero and would have distance one from P . By Lemma 2.1 and (2.5), each configuration in X is again the frequency set of an exponential Riesz basis. Only the uniform projection estimate will be needed below.

3.3. Translation averaging. Put $C_R = [-R, R]^n$ and define a probability measure on X by

$$\mu_R = \frac{1}{|C_R|} \int_{C_R} \delta_{T_v \Lambda} dv.$$

Compactness gives a weakly convergent subsequence as $R \rightarrow \infty$; write μ for its limit (we don't claim it is unique). For $F \in C(X)$ and fixed y ,

$$\left| \int F \circ T_y d\mu_R - \int F d\mu_R \right| \leq \|F\|_\infty \frac{|(C_R + y) \triangle C_R|}{|C_R|} \rightarrow 0.$$

Passing to the limit shows that μ is invariant under every T_y . Let

$$\mathcal{H} = L^2(X, \mu), \quad (U_y f)(\Gamma) = f(\Gamma + y). \quad (3.3)$$

These operators form a unitary representation of $\mathbb{R}^n$. Joint continuity of the action on the compact space X makes it pointwise continuous on $C(X)$, and density extends this conclusion to $\mathcal{H}$. This Hilbert space is separable. The constant function $\mathbf{1}$ has norm one and is fixed by all the U_y .

4. BOCHNER MEASURES AND STATIONARY FOURIER PROJECTIONS

We construct projections on $\mathcal{H}$ that retain the stationary frequencies θ for which $t + \theta \in \Omega$. From now on assume in addition that $|\partial\Omega| = 0$. The spectral variables, like the translation parameters, belong to $\mathbb{R}^n$.

4.1. Scalar Fourier measures. For $f \in \mathcal{H}$, the continuous function $y \mapsto \langle U_y f, f \rangle$ is positive definite. Bochner's theorem [15, Section 1.4.3, p. 19] gives a unique finite positive Borel measure σ_f on $\mathbb{R}^n$ such that

$$\langle U_y f, f \rangle = \int_{\mathbb{R}^n} e^{2\pi i y \cdot \theta} d\sigma_f(\theta), \quad \sigma_f(\mathbb{R}^n) = \|f\|^2. \quad (4.1)$$

Uniqueness and the parallelogram identity give

$$\sigma_{f+g} + \sigma_{f-g} = 2\sigma_f + 2\sigma_g, \quad \sigma_{cf} = |c|^2 \sigma_f. \quad (4.2)$$

Since $U_y \mathbf{1} = \mathbf{1}$, uniqueness also gives

$$\sigma_{\mathbf{1}} = \delta_0. \quad (4.3)$$

The zero-frequency mass will ensure that the boundary crossing changes the projection nontrivially.

For $a \in L^1(\mathbb{R}^n)$ define

$$T_a f = \int_{\mathbb{R}^n} a(y) U_y f dy, \quad \phi_a(\theta) = \int_{\mathbb{R}^n} a(y) e^{2\pi i y \cdot \theta} dy.$$

The first integral is an $\mathcal{H}$ -valued integral for each f , bounded in norm by $\|a\|_1 \|f\|$. Expanding its squared norm and using (4.1) gives

$$\begin{aligned} \|T_a f\|^2 &= \iint a(y) \overline{a(w)} \langle U_{y-w} f, f \rangle dy dw \\ &= \int_{\mathbb{R}^n} |\phi_a(\theta)|^2 d\sigma_f(\theta). \end{aligned}$$

Fubini applies because $a \in L^1$ and σ_f is finite. The group law and invariance give, for $a, c \in L^1$,

$$\|T_a f\|^2 = \int |\phi_a|^2 d\sigma_f, \quad T_a T_c = T_{a*c}, \quad T_a^* = T_{a^*}, \quad (4.4)$$

where $a^*(y) = \overline{a(-y)}$. In particular $\|T_a\| \leq \|\phi_a\|_\infty$. Convolution of kernels corresponds to multiplication of symbols, and $\phi_{a^*} = \overline{\phi_a}$. If two kernels have the same symbol, their operators agree by the norm identity. These facts provide the scalar calculus used below.

4.2. One measure to control all exceptional sets. Choose a dense sequence $(h_j)_{j \geq 1}$ in the unit sphere of $\mathcal{H}$ and set

$$\sigma = \sum_{j=1}^{\infty} 2^{-j} \sigma_{h_j}. \quad (4.5)$$

This is a finite positive measure of mass one. Moreover,

$$\sigma(E) = 0 \implies \sigma_f(E) = 0 \quad (f \in \mathcal{H}). \quad (4.6)$$

Indeed approximate f by scalar multiples g of the h_j . Then $\sigma_g(E) = 0$ and (4.2) gives

$$\sigma_f(E) \leq 2\sigma_{f-g}(E) + 2\sigma_g(E) \leq 2\|f - g\|^2 \longrightarrow 0.$$

This control measure allows a single choice of boundary point to work simultaneously for all vectors.

4.3. Continuous Fejér filters. The normalized overlap of the boxes $C_R = [-R, R]^n$ is

$$w_R(y) = \frac{|C_R \cap (C_R - y)|}{|C_R|} = \prod_{j=1}^n \left(1 - \frac{|y_j|}{2R}\right)_+. \quad (4.7)$$

Put

$$K_R(x) = \frac{1}{|C_R|} \left| \int_{C_R} e^{2\pi i v \cdot x} dv \right|^2, \quad m_R = K_R * \mathbf{1}_\Omega.$$

Plancherel gives $K_R \geq 0$ and $\int K_R = 1$. The scaling identity $K_R(x) = R^n K_1(Rx)$ shows that for each $\varepsilon > 0$, $\int_{|x| > \varepsilon} K_R(x) dx \rightarrow 0$. Thus K_R is an approximate identity. Since $\mathbf{1}_\Omega$ is locally constant away from its boundary,

$$0 \leq m_R \leq 1, \quad m_R(x) \longrightarrow \mathbf{1}_\Omega(x) \quad (x \notin \partial\Omega). \quad (4.8)$$

Expanding the square in K_R shows that $K_R = \mathcal{F}w_R$. Consequently, with p as in (2.1),

$$m_R(x) = \int_{\mathbb{R}^n} w_R(y) p(y) e^{2\pi i y \cdot x} dy. \quad (4.9)$$

This identity follows by Fubini: w_R has compact support and p is bounded. The same overlap weight will appear in the averaged inner products in Section 5.

Define

$$\Pi_{t,R}f = \int_{\mathbb{R}^n} w_R(y)p(y)e^{2\pi it \cdot y} U_y f dy. \quad (4.10)$$

This is T_a for an integrable kernel with symbol $m_R(t + \theta)$. By (4.4) and (4.8), each $\Pi_{t,R}$ is a self-adjoint contraction.

4.4. Constructing the cutoff projections.

Proposition 4.1. *The set*

$$\mathcal{G} = \{t \in \mathbb{R}^n : \sigma(\partial\Omega - t) = 0\} \quad (4.11)$$

is measurable and has full Lebesgue measure. For every $t \in \mathcal{G}$, $\Pi_{t,R}$ converges pointwise as $R \rightarrow \infty$ to an orthogonal projection Π_t . For $s, t \in \mathcal{G}$ and $f \in \mathcal{H}$,

$$\|\Pi_t f\|^2 = \int_{\mathbb{R}^n} \mathbf{1}_\Omega(t + \theta) d\sigma_f(\theta), \quad (4.12)$$

$$\|(\Pi_t - \Pi_s)f\|^2 = \int_{\mathbb{R}^n} |\mathbf{1}_\Omega(t + \theta) - \mathbf{1}_\Omega(s + \theta)|^2 d\sigma_f(\theta). \quad (4.13)$$

Proof. The function $(t, \theta) \mapsto \mathbf{1}_{\partial\Omega}(t + \theta)$ is Borel measurable, so its integral against σ is measurable. Tonelli and the boundary-null assumption give

$$\int_{\mathbb{R}^n} \sigma(\partial\Omega - t) dt = \int_{\mathbb{R}^n} |\partial\Omega - \theta| d\sigma(\theta) = 0. \quad (4.14)$$

Thus $\mathcal{G}$ is conull. Fix $t \in \mathcal{G}$. By (4.6), the boundary exception is null for every σ_f . The difference norm formula for the integrable kernels gives

$$\|(\Pi_{t,R} - \Pi_{t,S})f\|^2 = \int |m_R(t + \theta) - m_S(t + \theta)|^2 d\sigma_f(\theta).$$

The bounded symbols converge in $L^2(\sigma_f)$ by dominated convergence. Hence $\Pi_{t,R}f$ is Cauchy and has a limit $\Pi_t f$. Uniform boundedness makes Π_t a bounded linear contraction. Similarly,

$$\|(\Pi_{t,R}^2 - \Pi_{t,R})f\|^2 = \int |m_R(t + \theta)^2 - m_R(t + \theta)|^2 d\sigma_f(\theta) \rightarrow 0.$$

Products pass to the pointwise limit for uniformly bounded operators:

$$\|\Pi_{t,R}^2 f - \Pi_t^2 f\| \leq \|\Pi_{t,R}\| \|\Pi_{t,R}f - \Pi_t f\| + \|(\Pi_{t,R} - \Pi_t)\Pi_t f\| \rightarrow 0.$$

Thus $\Pi_t^2 = \Pi_t$. Self-adjointness passes through inner products, so Π_t is orthogonal. Passing to limits in the norm identities for $\Pi_{t,R}$ and $\Pi_{t,R} - \Pi_{s,R}$ proves the asserted formulas. $\square$

We call $\theta \mapsto \mathbf{1}_\Omega(t + \theta)$ the cutoff symbol of Π_t . Here t is a parameter and θ is a frequency of the stationary translation action. Formula (4.12) says that $\|\Pi_t f\|^2$ is the spectral mass of f in $\Omega - t$. Formula (4.13) will construct the two boundary limits.

5. PRESERVING THE DISTANCE BOUND

We transfer (3.2) to $\mathcal{H}$ by averaging scalar inner products of compactly supported functions on $\mathbb{R}^n$. The comparison operator is now a family M_t . We first prove that its members are orthogonal projections and that this family is continuous in operator norm.

5.1. The covariant bump kernel. The projection Q_Γ has the continuous kernel

$$q_\Gamma(v, w) = \sum_{\lambda \in \Gamma} b(v - \lambda) \overline{b(w - \lambda)}. \quad (5.1)$$

For fixed v at most one summand is nonzero. Thus, uniformly in Γ ,

$$|q_\Gamma(v, w)| \leq \|b\|_\infty^2, \quad q_\Gamma(v, w) = 0 \quad \text{if } |v - w| > 2r. \quad (5.2)$$

Local finiteness, weak convergence and continuity of b imply joint continuity in (Γ, v, w) . The kernel satisfies

$$q_{\Gamma+y}(v, w) = q_\Gamma(v - y, w - y), \quad (5.3)$$

$$q_\Gamma(w, v) = \overline{q_\Gamma(v, w)}, \quad (5.4)$$

$$\int q_\Gamma(v, u) q_\Gamma(u, w) du = q_\Gamma(v, w). \quad (5.5)$$

The last identity follows by expanding the locally finite sums and using orthonormality of the bumps. Compact support makes the integrals absolute.

Set $k_y(\Gamma) = q_\Gamma(0, -y)$, and write B_y for multiplication by k_y on $\mathcal{H}$. The map $y \mapsto k_y$ is continuous with values in $C(X)$: joint continuity is uniform on compact subsets of $X \times \mathbb{R}^n$. In particular $y \mapsto B_y U_y f$ is continuous for every $f \in \mathcal{H}$. Moreover it vanishes for $|y| > 2r$. Define

$$M_t f = \int_{\mathbb{R}^n} e^{2\pi i t \cdot y} B_y U_y f dy. \quad (5.6)$$

This is an $\mathcal{H}$ -valued integral with norm at most $\|f\| \int \|k_y\|_\infty dy$.

Proposition 5.1. *Every M_t is an orthogonal projection. The family is Lipschitz continuous in operator norm:*

$$\|M_t - M_s\| \leq L|t - s|, \quad L = 2\pi \int_{\mathbb{R}^n} |y| \|k_y\|_\infty dy < \infty. \quad (5.7)$$

Proof. For bounded f, g , invariance of μ and the substitution $\Gamma' = \Gamma + y$ give

$$\begin{aligned} \langle B_y U_y f, g \rangle &= \int_X q_\Gamma(0, -y) f(\Gamma + y) \overline{g(\Gamma)} d\mu(\Gamma) \\ &= \int_X q_{\Gamma'}(y, 0) f(\Gamma') \overline{g(\Gamma' - y)} d\mu(\Gamma') \\ &= \langle f, B_{-y} U_{-y} g \rangle. \end{aligned}$$

Density gives $(B_y U_y)^* = B_{-y} U_{-y}$. Taking the adjoint in (5.6) and changing y to $-y$ therefore gives $M_t^* = M_t$.

For idempotence, covariance implies $k_u(\Gamma + y) = q_\Gamma(-y, -y - u)$. First let f be bounded. Expanding M_t^2 and putting $v = y + u$ yields

$$\begin{aligned} (M_t^2 f)(\Gamma) &= \int e^{2\pi i t \cdot v} \left[\int q_\Gamma(0, -y) q_\Gamma(-y, -v) dy \right] f(\Gamma + v) dv \\ &= \int e^{2\pi i t \cdot v} q_\Gamma(0, -v) f(\Gamma + v) dv = (M_t f)(\Gamma), \end{aligned}$$

where the second equality is (5.5), with the change of variable $u = -y$. The original double integral has $|y|, |u| \leq 2r$ and uniformly bounded coefficients, so Fubini applies. The equality extends to every $f \in \mathcal{H}$ by density and boundedness. Hence $M_t^2 = M_t$, and M_t is an orthogonal projection.

Finally $|e^{2\pi i t \cdot y} - e^{2\pi i s \cdot y}| \leq 2\pi |t - s| |y|$. Applying the triangle inequality to the vector integral proves (5.7). Finiteness follows from (5.2). $\square$

The norm-continuity comes from the integrable, compactly supported kernel. The translation operators themselves are only required to be pointwise continuous. This distinction will matter at the boundary crossing.

5.2. Compactly supported functions from a stationary vector.

Fix $R > 0$, $t \in \mathbb{R}^n$, and abbreviate $C = C_R$. For $f \in \mathcal{H}$ define, for almost every Γ ,

$$a_f(\Gamma)(v) = |C|^{-1/2} \mathbf{1}_C(v) e^{-2\pi i t \cdot v} f(\Gamma - v). \quad (5.8)$$

These are compactly supported functions of $v \in \mathbb{R}^n$. Tonelli and stationarity give

$$\int_X \|a_f(\Gamma)\|_2^2 d\mu(\Gamma) = \frac{1}{|C|} \int_C \int_X |f(\Gamma - v)|^2 d\mu(\Gamma) dv = \|f\|^2. \quad (5.9)$$

In particular $a_f(\Gamma) \in L^2(\mathbb{R}^n)$ for almost every Γ . Borel representatives of f give joint measurability, and the same identity applied to a representative that vanishes almost everywhere shows that the resulting integrals do not depend on representatives. Thus the uncountable set of translation parameters causes no null-set ambiguity.

5.3. Compute the two averaged inner products. Let

$$M_{t,R} f = \int w_R(y) e^{2\pi i t \cdot y} B_y U_y f dy. \quad (5.10)$$

For bounded f, g , expansion of the compact spatial integrals gives

$$\begin{aligned} &\int_X \langle Q_\Gamma a_f(\Gamma), a_g(\Gamma) \rangle d\mu(\Gamma) \\ &= \frac{1}{|C|} \int_{C \times C} e^{2\pi i t \cdot (v-w)} \int_X q_\Gamma(v, w) f(\Gamma - w) \overline{g(\Gamma - v)} d\mu(\Gamma) dv dw. \end{aligned}$$

In the inner integral substitute $\Gamma' = \Gamma - v$. For $y = v - w$, covariance turns the kernel into $q_{\Gamma'}(0, -y)$, and the two functions become $f(\Gamma' + y)$ and $g(\Gamma')$. The inner integral is therefore $\langle B_y U_y f, g \rangle$. For every integrable function H of the difference variable,

$$\frac{1}{|C|} \int_{C \times C} H(v - w) dv dw = \int w_R(y) H(y) dy.$$

Consequently

$$\int_X \langle Q_\Gamma a_f, a_g \rangle d\mu = \langle M_{t,R} f, g \rangle. \quad (5.11)$$

For P its kernel $p(v - w)$ is independent of Γ . The same calculation, now with inner integral $\langle U_{v-w} f, g \rangle$, gives

$$\begin{aligned} \int_X \langle P a_f, a_g \rangle d\mu &= \int w_R(y) p(y) e^{2\pi i t \cdot y} \langle U_y f, g \rangle dy \\ &= \langle \Pi_{t,R} f, g \rangle. \end{aligned} \quad (5.12)$$

All kernels are bounded on $C \times C$. Thus the expansions for bounded f, g are justified by absolute integration. For general $f, g \in \mathcal{H}$, Cauchy–Schwarz, contractivity of P, Q_Γ and (5.9) bound each averaged form by $\|f\| \|g\|$. The identities extend by density.

5.4. Apply the distance bound and pass to the limit.

Proposition 5.2. *For every $t \in \mathcal{G}$,*

$$\|\Pi_t - M_t\| \leq \gamma < 1. \quad (5.13)$$

Proof. Subtracting (5.11) from (5.12) gives

$$\langle (\Pi_{t,R} - M_{t,R}) f, g \rangle = \int_X \langle (P - Q_\Gamma) a_f(\Gamma), a_g(\Gamma) \rangle d\mu. \quad (5.14)$$

Apply (3.2) for almost every Γ , followed by Cauchy–Schwarz in X and (5.9):

$$\begin{aligned} |\langle (\Pi_{t,R} - M_{t,R}) f, g \rangle| &\leq \gamma \int_X \|a_f(\Gamma)\|_2 \|a_g(\Gamma)\|_2 d\mu \\ &\leq \gamma \|f\| \|g\|. \end{aligned}$$

Thus $\|\Pi_{t,R} - M_{t,R}\| \leq \gamma$ for every t and R . By (5.2) and dominated convergence,

$$\|M_{t,R} - M_t\| \leq \int_{|y| \leq 2r} |1 - w_R(y)| \|k_y\|_\infty dy \longrightarrow 0, \quad (5.15)$$

uniformly in t . For $t \in \mathcal{G}$, Proposition 4.1 gives pointwise convergence $\Pi_{t,R} \rightarrow \Pi_t$. Testing the distance bound on each vector and taking the limit proves (5.13). $\square$

The finite averages $M_{t,R}$ need not be projections. Their norm limit M_t is a projection by Proposition 5.1. At a boundary parameter, the spectral cutoffs will have two different pointwise limits, whereas M_t has a single operator-norm limit. This gives a common comparison projection for the two sides.

6. CROSSING THE SPHERE

We now prove Theorem 1.1. Suppose that a ball $\Omega = B(a, \rho) \subset \mathbb{R}^n$, $n \geq 2$, admits an exponential Riesz basis. The preceding sections give projections Π_t and M_t with (5.13) for $t \in \mathcal{G}$. We will find two boundary limits whose ranges satisfy a strict inclusion, both within distance $\gamma < 1$ of M_{t_0} .

Recall that Π_t retains the stationary frequencies θ for which $t + \theta \in \Omega$. We approach a boundary point t_0 from outside and inside. At $\theta = 0$, this changes the cutoff from zero to one. The geometric step chooses t_0 so that every other frequency on the translated boundary lies in a common spectral null set.

6.1. Choose a boundary point with no other spectral boundary mass. Write $S = \partial\Omega$ and $\nu = \mathcal{H}^{n-1}|_S$. This is a finite nonzero measure.

Lemma 6.1. *For every nonzero $\theta \in \mathbb{R}^n$,*

$$\nu(S \cap (S - \theta)) = 0. \quad (6.1)$$

Moreover, S has Lebesgue measure zero.

Proof. If $x \in S \cap (S - \theta)$, then $|x - a|^2 = \rho^2$ and $|x + \theta - a|^2 = \rho^2$. Subtraction gives

$$2(x - a) \cdot \theta = -|\theta|^2.$$

For $\theta \neq 0$, this is a hyperplane. Its intersection with the sphere is empty, a point, or an $(n - 2)$ -dimensional sphere, hence has zero $\mathcal{H}^{n-1}$ -measure. Finally a sphere in $\mathbb{R}^n$ has zero n -dimensional Lebesgue measure. $\square$

We need the reverse viewpoint: fix a boundary point t_0 and control the frequencies θ for which $t_0 + \theta \in S$. Fubini interchanges these roles. Using the control measure σ from (4.5), we obtain

$$\begin{aligned} \int_S \sigma((S - t) \setminus \{0\}) d\nu(t) &= \int_{\mathbb{R}^n \setminus \{0\}} \nu(S \cap (S - \theta)) d\sigma(\theta) \\ &= 0. \end{aligned} \quad (6.2)$$

The integrand on the left is nonnegative and $\nu(S) > 0$. Thus we can choose $t_0 \in S$ such that

$$\sigma((S - t_0) \setminus \{0\}) = 0. \quad (6.3)$$

By (4.6), the same set is null for every σ_f . This is why we introduced one control measure: the chosen point works for every vector in $\mathcal{H}$ simultaneously.

Zero must be excluded from this null-set statement. Indeed, $\sigma_1 = \delta_0$, so the constant vector has spectral mass there. This mass is precisely what will enter the cutoff.

6.2. Approach the point from outside and inside. The good set $\mathcal{G}$ has full Lebesgue measure in $\mathbb{R}^n$, but this does not imply that it contains t_0 . In fact $t_0 \notin \mathcal{G}$: since $0 \in S - t_0$ and $\sigma_1 = \delta_0$, condition (4.11) cannot hold there. We therefore approach t_0 through good parameters and take limits of their projections.

Every neighborhood of a point on the sphere meets both Ω and its strict exterior in nonempty open sets. Each has positive Lebesgue measure, so each meets $\mathcal{G}$. We can consequently choose

$$t_j^- \in \mathcal{G} \cap (\mathbb{R}^n \setminus \overline{\Omega}), \quad t_j^+ \in \mathcal{G} \cap \Omega, \quad t_j^\pm \longrightarrow t_0.$$

These choices use open neighborhoods in the ambient space. There is no need to restrict the good parameters to a prescribed radial line.

Write the corresponding cutoff symbols as

$$a_j^\pm(\theta) = \mathbf{1}_\Omega(t_j^\pm + \theta).$$

If $t_0 + \theta$ is away from S , membership in Ω is locally constant, so both symbols eventually equal $\mathbf{1}_\Omega(t_0 + \theta)$. At $\theta = 0$, the exterior symbol is always zero and the interior symbol is always one. All remaining cases lie in $(S - t_0) \setminus \{0\}$, which is null for every σ_f by (6.3). Thus, for every $f \in \mathcal{H}$, we have $a_j^\pm \rightarrow m_\pm$ σ_f -almost everywhere, where

$$m_-(\theta) = \mathbf{1}_{\Omega - t_0}(\theta), \tag{6.4}$$

$$m_+(\theta) = \mathbf{1}_{\Omega - t_0}(\theta) + \mathbf{1}_{\{0\}}(\theta). \tag{6.5}$$

Since Ω is open and $t_0 \in S$, zero does not belong to $\Omega - t_0$. Both limits are therefore indicator functions, and $m_- \leq m_+$. The only added spectral component is at zero.

6.3. Turn the limiting symbols into projections. We now justify that the projections themselves converge. For either choice of sign, (4.13) gives

$$\|(\Pi_{t_j^\pm} - \Pi_{t_k^\pm})f\|^2 = \int_{\mathbb{R}^n} |a_j^\pm - a_k^\pm|^2 d\sigma_f.$$

The right-hand side tends to zero, since

$$|a_j^\pm - a_k^\pm|^2 \leq 2|a_j^\pm - m_\pm|^2 + 2|a_k^\pm - m_\pm|^2$$

and dominated convergence applies to both terms. Hence $(\Pi_{t_j^\pm} f)_j$ is Cauchy for each f . The uniform bound $\|\Pi_t\| \leq 1$ gives bounded linear operators R_-, R_+ with

$$\Pi_{t_j^-} \longrightarrow R_-, \quad \Pi_{t_j^+} \longrightarrow R_+ \quad \text{pointwise.} \tag{6.6}$$

They are orthogonal projections. To see the idempotence explicitly, if $P_j = \Pi_{t_j^\pm}$ and $R = R_\pm$, then

$$\|P_j^2 f - R^2 f\| \leq \|P_j\| \|P_j f - Rf\| + \|(P_j - R)Rf\| \longrightarrow 0.$$

Since $P_j^2 = P_j$, we obtain $R^2 = R$. Self-adjointness passes to the limit through inner products. Finally, passing to the limit in (4.12) gives

$$\|R_\pm f\|^2 = \int_{\mathbb{R}^n} m_\pm(\theta) d\sigma_f(\theta). \quad (6.7)$$

This formula lets us read the order of the limiting projections from the order of their symbols.

6.4. Strict inclusion of subspaces and the contradiction. Since $m_- \leq m_+$, (6.7) implies $\|R_- f\| \leq \|R_+ f\|$ for every f . In particular, $R_+ f = 0$ implies $R_- f = 0$. Thus

$$\ker R_+ \subset \ker R_-, \quad \text{Ran } R_- \subset \text{Ran } R_+,$$

where the second inclusion follows by taking orthogonal complements.

To see that this inclusion is strict, use the constant vector. Its spectral measure is δ_0 , so (6.7) gives

$$R_- \mathbf{1} = 0, \quad \|R_+ \mathbf{1}\| = \|\mathbf{1}\| = 1.$$

An orthogonal projection preserves a vector's norm only when it fixes that vector, by Pythagoras. Hence $R_+ \mathbf{1} = \mathbf{1}$: the constant function belongs to the larger range and is orthogonal to the smaller one. We have proved

$$\text{Ran } R_- \subsetneq \text{Ran } R_+. \quad (6.8)$$

More precisely, the difference is an orthogonal projection, and

$$\|(R_+ - R_-)f\|^2 = \sigma_f(\{0\}), \quad (R_+ - R_-)\mathbf{1} = \mathbf{1}. \quad (6.9)$$

The jump need not have rank one; the constant vector ensures that it is nonzero.

Both limits also inherit a common distance bound. By Proposition 5.1, $M_{t_j^\pm} \rightarrow M_{t_0}$ in operator norm. For every $f \in \mathcal{H}$,

$$\begin{aligned} \|(R_\pm - M_{t_0})f\| &= \lim_j \|(\Pi_{t_j^\pm} - M_{t_0})f\| \\ &\leq \limsup_j \left(\gamma + \|M_{t_j^\pm} - M_{t_0}\| \right) \|f\| = \gamma \|f\|. \end{aligned}$$

Consequently

$$\|R_- - M_{t_0}\| \leq \gamma < 1, \quad \|R_+ - M_{t_0}\| \leq \gamma < 1. \quad (6.10)$$

The comparison at t_0 is legitimate even though $t_0 \notin \mathcal{G}$: M_t is defined and is a projection for every real parameter. Lemma 2.2 contradicts (6.8) and (6.10). This proves Theorem 1.1.

Remark 6.2 (The role of the stationary space). The invariant probability measure supplies a nonzero constant vector with spectral measure δ_0 . The ordinary translation representation on $L^2(\mathbb{R}^n)$ has no such vector. Its scalar spectral measures are $|\mathcal{F}a(\theta)|^2 d\theta$, so changing a cutoff at the single point zero has no effect on that space. On $\mathcal{H}$, the same change gives $R_- \mathbf{1} = 0$ and $R_+ \mathbf{1} = \mathbf{1}$. The averaging argument brings both this invariant vector and the original distance bound into the same stationary space.

Remark 6.3. For a sphere the boundary has no positive-measure overlap with a nontrivial translate. This fails for the one-dimensional interval: a translation can send one endpoint to the other. It also fails for opposite sides of a rectangle. A boundary crossing can then exchange spectral components, and the contradiction from strict inclusion does not follow.

Remark 6.4 (A more general boundary condition). Let $\Omega \subset \mathbb{R}^n$ be a nonempty bounded open set with $|\partial\Omega| = 0$. Suppose its boundary carries a finite nonzero positive Borel measure ν such that

$$\nu(\partial\Omega \cap (\partial\Omega - \theta)) = 0 \quad (\theta \neq 0).$$

Assume also that, for ν -almost every $t \in \partial\Omega$, every neighborhood of t meets both Ω and $\mathbb{R}^n \setminus \bar{\Omega}$ in sets of positive Lebesgue measure. Then $L^2(\Omega)$ admits no Riesz basis of exponentials. Indeed Sections 2–5 apply under these hypotheses. Tonelli with ν supplies a point satisfying both the spectral boundary-null condition and the local separation condition. The latter supplies the two sequences of good parameters. Their symbols have exactly the limits (6.4)–(6.5), and the same arguments using pointwise limits and norm continuity give a contradiction. In particular one may use surface measure whenever it is finite and nonzero and satisfies the displayed overlap and separation conditions.

7. TRIANGLES

For a triangle, translations along an edge can overlap it in positive length. The other edges are not parallel to it, so all surviving boundary components can still be made to enter the cutoff together.

Theorem 7.1. *No nondegenerate triangle in $\mathbb{R}^2$ admits a Riesz basis of exponentials.*

Proof. An invertible affine change of variables preserves the Riesz basis property: frequencies are transformed by the transpose of the linear part, and the change introduces only unit phases and a fixed norm factor. Thus we may assume

$$\Omega = \{x_1 > 0, x_2 > 0, x_1 + x_2 < 1\}.$$

Suppose it has an exponential Riesz basis. Since Ω is bounded and its boundary has area zero, Sections 2–5 give the control measure σ , the conull set $\mathcal{G}$, and the projections with (5.13).

Let $L = \{(a, 0) : 0 < a < 1\}$ and $F = \partial\Omega \setminus L$. The set F consists of the two other closed edges, including all vertices. For $\nu = \mathcal{H}^1|_L$ and every $\theta \in \mathbb{R}^2$,

$$\nu(L \cap (F - \theta)) = 0 :$$

a horizontal segment meets a translated vertical or diagonal supporting line in at most one point. Tonelli therefore gives

$$\int_L \sigma(F - t) d\nu(t) = \int_{\mathbb{R}^2} \nu(L \cap (F - \theta)) d\sigma(\theta) = 0.$$

Choose $t_0 \in L$ with $\sigma(F - t_0) = 0$. This set is null for every σ_f by (4.6).

Near t_0 , the triangle coincides with the half-plane $x_2 > 0$. Choose good parameters $t_j^+ \in \mathcal{G} \cap \Omega$ and $t_j^- \in \mathcal{G} \cap (\mathbb{R}^2 \setminus \overline{\Omega})$ tending to t_0 inside such a neighborhood. Their second coordinates are positive and negative, respectively. For each point $t_0 + \theta \in L$, membership in the triangle is likewise locally exactly positivity of the second coordinate. Therefore $\mathbf{1}_\Omega(t_j^+ + \theta) \rightarrow 1$ and $\mathbf{1}_\Omega(t_j^- + \theta) \rightarrow 0$ on $L - t_0$. Away from the boundary both symbols converge to $\mathbf{1}_\Omega(t_0 + \theta)$. All other boundary points belong to the common null set $F - t_0$.

The limiting symbols are consequently

$$m_- = \mathbf{1}_{\Omega - t_0}, \quad m_+ = \mathbf{1}_{\Omega - t_0} + \mathbf{1}_{L - t_0}.$$

They are indicators of sets ordered by inclusion. Formula (4.13) and dominated convergence make $(\Pi_{t_j^\pm} f)_j$ Cauchy for every f . As in Section 6, the resulting pointwise limits R_-, R_+ are orthogonal projections with

$$\|R_\pm f\|^2 = \int m_\pm d\sigma_f.$$

Thus $\|R_- f\| \leq \|R_+ f\|$ for every f , which gives $\text{Ran } R_- \subset \text{Ran } R_+$ by taking kernels and orthogonal complements. Since $0 \in L - t_0$ and $\sigma_1 = \delta_0$, we have $R_- \mathbf{1} = 0$ and $R_+ \mathbf{1} = \mathbf{1}$; the inclusion is strict. Finally $M_{t_j^\pm} \rightarrow M_{t_0}$ in norm, so (5.13) gives $\|R_\pm - M_{t_0}\| \leq \gamma < 1$, contradicting Lemma 2.2.

Here $\|(R_+ - R_-)f\|^2 = \sigma_f(L - t_0)$. The jump may contain nonzero tangential frequencies; the constant vector guarantees that it is nonzero. $\square$

Remark 7.2 (Convex polygons with an unpaired edge). The same proof excludes an exponential Riesz basis on the interior of a convex polygon having an edge with no parallel partner. Edges are taken to be maximal boundary segments. In particular the conclusion holds for every convex polygon with an odd number of sides: a bounded convex polygon has at most two maximal edges in each unoriented direction, so an odd number of sides cannot all be paired by parallelism.

To verify the extension, let L be the relative interior of an unpaired edge and put $F = \partial\Omega \setminus L$. Every other edge is nonparallel to L , so for

$$\nu = \mathcal{H}^1|_L, \quad \nu(L \cap (F - \theta)) = 0 \quad (\theta \in \mathbb{R}^2).$$

The intersections with the translated supporting lines are finite; the two endpoints contribute only finitely many further points. Tonelli gives $t_0 \in L$ with $\sigma(F - t_0) = 0$. Fix the inward unit normal to L . At each point of L , membership in Ω is locally determined by the positive sign of the displacement in this normal direction. Choose good parameters approaching t_0 from the two local half-planes. For every $t_0 + \theta \in L$, the same normal sign determines their eventual membership. The symbols are therefore the same m_-, m_+ as in the triangle proof, with this edge L . Their limiting projections satisfy $\text{Ran } R_- \subsetneq \text{Ran } R_+$ because $0 \in L - t_0$. Norm-continuity of M_t again supplies the common comparison projection M_{t_0} and Lemma 2.2 gives the contradiction.

8. ELLIPSOIDS

Corollary 8.1. *For every $n \geq 2$, no nondegenerate ellipsoid in $\mathbb{R}^n$ admits a Riesz basis of exponentials.*

Proof. Write the ellipsoid as $\Omega = a + AB(0, 1)$, where A is an invertible real $n \times n$ matrix. The pullback $f \mapsto [y \mapsto f(a + Ay)]$ is an isomorphism from $L^2(\Omega)$ onto $L^2(B(0, 1))$ with norm factor $|\det A|^{-1/2}$. It sends e_λ to $e^{2\pi i \lambda \cdot a} e_{A^\top \lambda}$. The invertible change of frequency and the unit phases preserve the Riesz basis property. A basis on Ω would therefore give one on the unit ball, contradicting Theorem 1.1. $\square$

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